CHAPTER 3

EVOLUTION

CHAPTER ESSENCE AT A GLANCE:
Prebiotic chemical origins, comparative anatomical evidence, natural selection dynamics, and Hardy-Weinberg equilibrium.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Chemical Origin of Life and Miller's Simulation:** Oparin-Haldane theory proposed that life originated abiogenically from non-living organic molecules in a reducing primitive atmosphere. Stanley Miller (1953) simulated prebiotic Earth conditions: spark discharge chamber containing CH4, NH3, H2O vapor, and H2 at 800°C; synthesized amino acids (glycine, alanine, aspartic acid), experimentally confirming chemical abiogenesis.

• **Homology versus Analogy and Adaptive Radiation:** Homologous organs share common anatomical origin and basic structure but perform different functions (divergent evolution, e.g., forelimbs of whales, bats, cheetah, and humans). Analogous organs share similar functions adapted to similar ecological niches but differ in anatomical origin (convergent evolution, e.g., wings of butterflies and birds). Adaptive radiation: evolutionary diversification from an ancestral stock into diverse ecological niches (Darwin's finches, Australian marsupials).

• **Natural Selection Types on Phenotypic Traits:** Natural selection acts on phenotypic distributions across three modes: (1) Stabilizing selection: favors average intermediate phenotypes, making the bell curve narrower; (2) Directional selection: favors one extreme phenotype, shifting the peak in one direction (industrial melanism in peppered moth Biston betularia); (3) Disruptive selection: favors both extreme phenotypes over the intermediate, splitting the peak into two.

• **Hardy-Weinberg Equilibrium and Human Ancestry:** Hardy-Weinberg principle states that allele frequencies in a large, randomly mating population remain constant from generation to generation in the absence of evolutionary forces: $p^2 + 2pq + q^2 = 1$ ($p + q = 1$). Disrupted by gene flow, genetic drift (Founder and Bottleneck effects), mutation, recombination, and natural selection. Human lineage: Australopithecus (bipedal, 500 cc), Homo habilis (first toolmaker, 650-800 cc), Homo erectus (meat-eater, 900 cc), Neanderthal man (buried dead, 1400 cc), and Homo sapiens.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$(p+q)^2 = p^2 + 2pq + q^2 = 1 \quad [p = \text{freq}(A),\ q = \text{freq}(a)]$$

Hardy-Weinberg Binomial Population Expansion

$$\text{Prebiotic Spark Mix} = CH_4 + NH_3 + H_2O + H_2 \xrightarrow{800^\circ C,\ \text{Electric Spark}} \text{Amino Acids}$$

Stanley Miller Prebiotic Abiogenic Synthesis

$$\text{Evolutionary Trajectories} = \frac{\text{Homology: Divergent (Common origin, different function)}}{\text{Analogy: Convergent (Different origin, similar function)}}$$

Comparative Anatomical Evolutionary Dichotomy

$$\text{Phenotypic Selection} = \text{Stabilizing (Peak narrows)} \Longleftrightarrow \text{Directional (Peak shifts)} \Longleftrightarrow \text{Disruptive (Two peaks)}$$

Modes of Natural Selection on Phenotypic Distributions

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Homologous organs reflect DIVERGENT evolution (shared common ancestry), whereas Analogous organs reflect CONVERGENT evolution (adaptation to similar ecological demands).

• **Alert:** In Hardy-Weinberg equilibrium, p is the frequency of the dominant allele, q is the frequency of the recessive allele, p^2 is homozygous dominant, 2pq is heterozygous, and q^2 is homozygous recessive.

• **Alert:** Homo habilis had a cranial capacity of 650 to 800 cc and did not eat meat; Homo erectus had 900 cc and ate meat; Neanderthal man had 1400 cc.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between homologous organs and analogous organs with an anatomical example for each in animals. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Homologous organs: Structures sharing a common anatomical origin and basic embryonic plan but adapted to perform different functions (divergent evolution, e.g., forelimbs of cheetah, whale, and human). **[1 Mark]**

• Analogous organs: Structures performing similar functions but possessing completely different anatomical plans and evolutionary origins (convergent evolution, e.g., wings of insects and birds). **[1 Mark]**

2. What is adaptive radiation? Describe how Darwin's finches on the Galapagos islands illustrate this evolutionary phenomenon. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Definition: The evolutionary process in which different species radiate from a common ancestral point and adapt to diverse ecological geographical niches. **[1 Mark]**

• Darwin's finches: An ancestral seed-eating finch stock radiated across isolated Galapagos islands, evolving diverse beak sizes and shapes suited for feeding on insects, seeds, and cactus nectar. **[1 Mark]**

3. Describe Stanley Miller's classic 1953 spark discharge experiment. Which gases were used and which organic molecules were synthesized? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Experimental apparatus: Closed glass circulation system with tungsten electrodes providing high-voltage electric spark discharges at 800 degrees Celsius to simulate prebiotic lightning. **[1 Mark]**

• Gaseous mixture: Methane (CH_4), Ammonia (NH_3), Hydrogen (H_2), and Water vapor (H_2O) simulating the primitive reducing atmosphere. **[1 Mark]**

• Synthesized products: After one week of continuous operation, Miller detected formation of simple organic amino acids: Glycine, Alanine, and Aspartic acid, validating chemical evolution. **[1 Mark]**

4. Explain the three types of natural selection (stabilizing, directional, and disruptive) using bell-shaped trait distribution curves. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Stabilizing selection: Favors intermediate mean phenotypes while eliminating extreme variations; the bell-shaped distribution curve becomes narrower and taller over generations. **[1 Mark]**

• Directional selection: Environmental change favors individuals at one extreme of the phenotypic distribution; the peak of the curve shifts in that specific direction (e.g., industrial melanism). **[1 Mark]**

• Disruptive selection: Environmental conditions favor individuals at both extremes over intermediate phenotypes; the central peak dips, forming two distinct peaks. **[1 Mark]**

5. In a human population at Hardy-Weinberg equilibrium, the frequency of an autosomal recessive allele causing cystic fibrosis is 0.2. Calculate the frequency of the dominant allele, the percentage of homozygous dominant individuals, and the percentage of heterozygous carriers. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Recessive allele frequency: $q = 0.2$. Dominant allele frequency: $p = 1 - q = 1 - 0.2 = 0.8$. **[1 Mark]**

• Homozygous dominant frequency: $p^2 = (0.8)^2 = 0.64 = 64$ percent of the population. **[1 Mark]**

• Heterozygous carrier frequency: $2pq = 2 * 0.8 * 0.2 = 0.32 = 32$ percent of the population. **[1 Mark]**

6. Trace the evolutionary lineage of modern humans through Australopithecus, Homo habilis, Homo erectus, and Homo neanderthalensis, specifying the cranial capacity of each. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Australopithecus: Lived in East African grasslands; walked upright, bipedal locomotion, cranial capacity around 500 cc. **[0.75 Mark]**

• Homo habilis: First human-like hominid (toolmaker); did not eat meat, cranial capacity 650 to 800 cc. **[0.75 Mark]**

• Homo erectus: Appeared around 1.5 million years ago; ate meat, possessed prominent brow ridges, cranial capacity about 900 cc. **[0.75 Mark]**

• Homo neanderthalensis: Lived in central Asia; used animal hides to protect bodies and buried their dead, cranial capacity about 1400 cc. **[0.75 Mark]**

7. Before industrialization in 1850s England, white-winged peppered moths (Biston betularia) were far more abundant than melanic dark-winged moths. By 1920, dark-winged moths predominated industrial regions. Explain this shift in terms of natural selection, and predict the population outcome if industrial air pollution is cleaned. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Pre-industrial condition: Tree trunks were covered with thick, light-colored lichens; white-winged moths were camouflaged against predatory birds, whereas rare dark melanic mutants were easily spotted and predated. **[1.5 Marks]**

• Post-industrial selection: Industrial smoke and soot killed pollution-sensitive lichens and blackened tree bark; now, dark melanic moths enjoyed camouflage, while white moths were easily detected and predated, shifting the population frequency (directional selection). **[1.5 Marks]**

• Clean air prediction: With strict anti-pollution legislation, soot disappears and lichens return; light-colored moths will regain camouflage advantage, and natural selection will reverse to favor the white-winged phenotype again. **[1 Mark]**